

Solving forward and inverse problems for atmospheric radio frequency discharge by Physics-Informed Neural Networks

Cite as: J. Appl. Phys. **139**, 083302 (2026); doi: [10.1063/5.0307593](https://doi.org/10.1063/5.0307593)

Submitted: 17 October 2025 · Accepted: 30 January 2026 ·

Published Online: 23 February 2026



Wenkai Li and Yuantao Zhang

AFFILIATIONS

School of Electrical Engineering, Shandong University, Jinan, Shandong Province 250061, People's Republic of China

^{a)}Author to whom correspondence should be addressed: ytzhang@sdu.edu.cn

ABSTRACT

With the requirement of intelligent control of atmospheric plasmas system, more Artificial Intelligence (AI) algorithms should be introduced into the field of plasma simulation. In this study, an innovative Physics-Informed Neural Networks (PINNs) framework for solving both forward and inverse problems in atmospheric radio frequency (RF) plasma is explored. For forward problems, the PINNs architecture with multi-scale feature is established by coupling Poisson's equation, continuity equations, and drift-diffusion approximation equations into the loss function, to successfully capture the key plasma characteristics, showing good agreement with results from fluid simulation by discretization methods. For inverse problems that usually cannot be solved by discretization methods, by incorporating additional simulated (or measured) data of electric field as constraints in loss function, PINNs can accurately infer the applied voltage with relative errors smaller than 1%. The influence of various sampling positions, number of sampling points, and noise on the inversion of discharge parameter by PINNs is also investigated. In this study, according to the computational data, this mesh-less approach of PINNs successfully solves the fluid equations without relying on discretization methods and also shows the ability to inverse the discharge parameters such as applied voltage, driving frequency, or electrode spacing, given the data of electric field or plasma density, offering novel methodologies and insights for the intelligent control of atmospheric plasma systems.

© 2026 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>). <https://doi.org/10.1063/5.0307593>

I. INTRODUCTION

In recent decades, atmospheric pressure plasmas have high reactivity and are not required for vacuum facilities, which are widely applied in frontier fields including energy conversion,¹ agricultural nitrogen fixation,^{2,3} material surface modification,⁴ and biomedicine.⁵ Atmospheric radio frequency (RF) discharges can provide large-volume, homogenous, and stable plasmas, which are well accepted as ideal atmospheric plasma sources.^{6–8} It is very meaningful to study atmospheric RF discharge plasma through experiments and numerical algorithms.^{6,9–11} The current mainstream numerical simulation methods include fluid models,² particle-in-cell (PIC) algorithms,¹² and global models,¹³ typically solved by finite difference,¹⁴ finite element,¹⁵ finite volume, and other discretization methods.¹⁶ However, multi-scale modeling of complex discharges causes computational load to surge, and spatial

dimension extension triggers the curse of dimensionality, while refined meshing for higher accuracy further escalates computational overhead.¹⁷ Traditional techniques, hampered by inherent limitations in high computational complexity, prolonged time consumption, substantial hardware resource occupation, and poor convergence stability, struggle to meet large-scale, high-precision simulation requirements.^{18,19}

With the explosive growth of computing resources over the past decade, artificial intelligence (AI) technology is booming and showing a significant impact on plasma science.^{20,21} In the field of plasma simulation, machine learning algorithm, especially deep neural networks (DNNs), has also attracted extensive attention.²¹ The DNNs were applied to discuss the discharge characteristics with very high computational efficiency in the dielectric barrier discharges driven and RF discharges.^{22,23} A data-driven alternative model was trained using data from fluid simulation to enable the

07 March 2026 07:10:56

efficient prediction of various discharge characteristics for complex CO₂ plasma.^{24,25} Even the kinetic effects such as the electron energy distribution function (EEDF) can be accurately and efficiently predicted by DNNs based on the computational data from PIC simulation.²⁶ A feed-forward DNN was constructed to solve partial differential equations in thermal plasmas, and the potential of DNNs for thermal plasma modeling was demonstrated by solving a one-dimensional arc decay model.²⁷ However, the data-driven method also has very essential limitations that require substantial simulation or experimental data, yet acquiring such large data is often very challenging.²⁸ To address this limitation of data requirement, Physics-Informed Neural Networks (PINNs) have been developed by incorporating the physical information to reduce the requirement of data.²⁹ Usually, PINNs benefit from automatic differentiation technology, which avoids spatial discretization and does not need complicated grid division even when the study area is irregular.³⁰ And PINNs naturally handle high-dimensional inputs through neural networks, which break the curse of dimensionality.^{31,32} In addition, PINN is a good candidate for solving inverse problems due to its ease of integrating sparse and noisy data.³³

In recent years, PINNs²⁹ emerged as an alternative method to solve the partial differential equation (PDE) in computational science and engineering, for example, in optics,^{34,35} fluid mechanics,^{36,37} solid mechanics,^{38,39} and biomedicine.^{40,41} In the field of plasma numerical simulation, PINNs have also attracted attentions due to their potential advantages. The electron velocity distribution function in weakly ionized plasma was obtained by directly solving the Boltzmann equation based on PINNs.⁴² A CS-PINN (Coefficient Subnet Physics-Informed Neural Networks) and RK-PINN (Runge Kutta Physics-Informed Neural Networks) computational framework was developed for thermal plasma numerical simulation,⁴³ effectively addressing the challenges of solving variable coefficients in plasma simulations and achieving convergence for long-time equations. An artificial neural network model was built to analyze scanning drift tube experiments through electron group mapping, with electron transport coefficients determined via the electron continuity equation.⁴⁴ The fluid model describing DC-driven discharges, considering only collision ionization reactions, was solved by PINNs, yielding temporal and spatial distributions of electron density, ion density, and electric field.⁴⁵ Together, these studies show that although PINNs have shown their effectiveness in solving single or multiple plasma physical equations, their application in truly coupled multi-physical field plasma systems is still an unsolved challenge in this field.

In this study, both forward and inverse problems in RF discharges are investigated by PINNs. For the forward problem, by given discharge parameter of applied voltage, PINNs coupled with boundary/initial conditions (BC/IC) to calculate the plasma characteristics such as plasma density and electric field. For the inverse problem, leveraging the simulated (or experimentally measured) data of the electric field, PINNs are employed to inversely infer the discharge parameter of the applied voltage while inherently enforcing BC/IC constraints. Notably, traditional discretization-based methods for fluid model have struggled to solve inverse problems, whereas PINNs offer a novel mesh-free paradigm that bridges data-driven learning and physical constraints, opening new avenues for

the solution of inverse problems in plasma simulation, and the proposed PINNs framework is robust to noise and can adapt to the potential uncertainty in noisy data, making it well suited for applications with incomplete or uncertain data. The rest of this paper is organized as follows. A full description of the PINNs framework is given in Sec. II, especially the nondimensional equations in the loss function of PINNs are discussed in detail. The numerical data from PINNs are shown in Sec. III, and for the forward problem, PINNs successfully exported the temporal and spatial profiles of plasma density and electric field. For the inverse problem, the discharge parameters of the applied voltage are successfully inferred using PINNs by incorporating the additional data of the electric field. Finally, a summary is concluded in Sec. IV.

II. DESCRIPTION OF NUMERICAL METHODS

A. Description of fluid model

Usually, a one-dimensional fluid model can be applied to describe the atmospheric pressure discharges.^{46,47} In the present model, the pure helium plasmas are generated and maintained between two parallel-plate electrodes. The production and loss of the various particles are given by the continuity equations, and the particles flux is calculated by the drift-diffusion equations with the electric field given by the Poisson equation,⁹ which is given as

$$\frac{\partial N_{ie}(x, t)}{\partial t} + \frac{\partial J_{ie}(x, t)}{\partial x} = S(x, t), \quad (1)$$

$$J_{ie}(x, t) = \pm \mu_{ie} E(x, t) N_{ie}(x, t) - D_{ie} \frac{\partial N_{ie}(x, t)}{\partial x}, \quad (2)$$

where the subscripts *i* and *e* represent the electrons and positive ions in the discharge space, respectively. μ and D refer to the coefficients of mobility and diffusion, respectively. $N(x, t)$ and $J(x, t)$ are the density and flux, respectively. And $S(x, t)$ represents the particle source term, including the generation and loss of charged particles. In order to emphasize the process of solving fluid model using PINNs, only direct collision ionization reactions are considered in this work. So the specific mathematical expression of source term is as follows:

$$S(x, t) = Ap\mu_e E(x, t) N_e(x, t) e^{-\frac{Rp}{E(x, t)}}, \quad (3)$$

where p represents the gas pressure and A and B are parameters that depend on the type of working gas.

The electric field $E(x, t)$ is calculated by Poisson's equation,⁴⁶

$$\frac{\partial E(x, t)}{\partial x} = \frac{q}{\epsilon} (N_i(x, t) - N_e(x, t)), \quad (4)$$

where q and ϵ represent the elementary charge and vacuum permittivity, respectively. The electric field is integrated over the electrode spacing to obtain the potential $\varphi(x, t)$ at electrode spacing,

$$\frac{\partial \varphi(x, t)}{\partial x} = -E(x, t). \quad (5)$$

07 March 2026 07:10:56

For the parallel-plate RF discharge configuration with one electrode grounded and the other one connected to an RF power source, the boundary conditions for potential $\varphi(x, t)$ are specified as

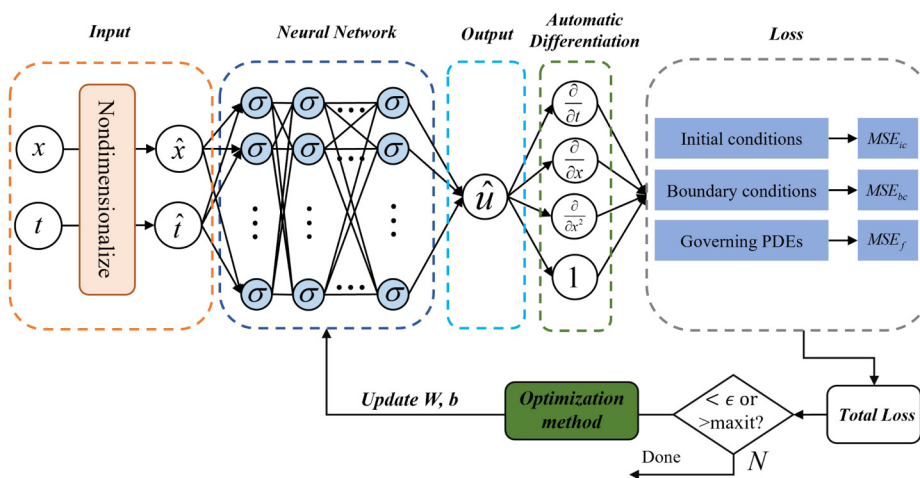
$$\varphi(0, t) = V_0 \sin(2\pi ft), \quad (6)$$

$$\varphi(d, t) = 0, \quad (7)$$

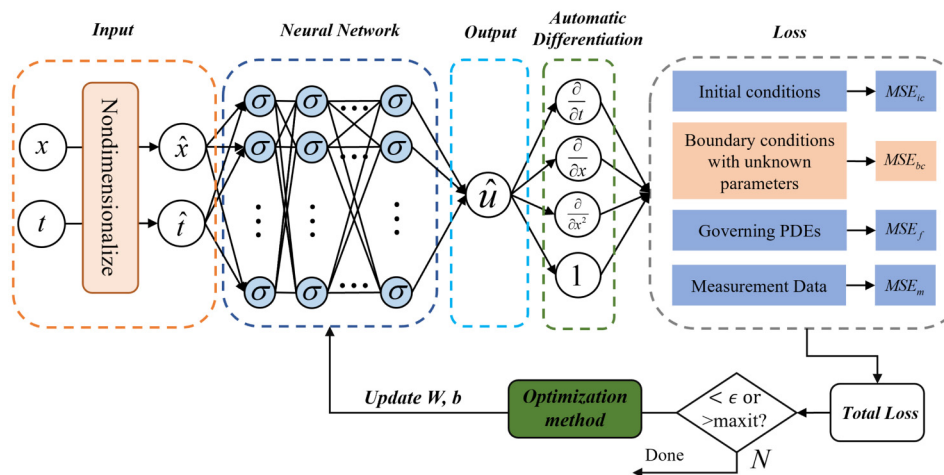
where V_0 is the amplitude of the applied voltage and f is the driving frequency of 13.56 MHz. d is the electrode spacing with a fixed value of 2.4 mm in this study.

B. PINNs for solving forward and inverse problems

PINNs can be seen as a universal function approximator based on compositional functions, which successfully integrate physical information with neural networks. During the training of neural networks, the PINNs are optimized to satisfy the forward and inverse PDE problems.^{29,48} The structures of the PINNs for solving the forward and inverse problems are shown in Fig. 1. The input coordinates (x, t) are dimensionless processed to eliminate the numerical instability caused by them; then the processed coordinates $(\hat{x}, \hat{t})$ are input into the neural network for forward propagation, and the output physical field prediction values $\hat{u}(\hat{x}, \hat{t})$ or unknown parameters λ are obtained by constraining of the loss function in the training process of PINNs. For the forward



(a) The PINNs architecture for the forward problem.



(b) The PINNs architecture for the inverse problem.

FIG. 1. The PINNs architecture for forward (a) and inverse (b) problems. For forward problems, the loss function comprises the nondimensional PDEs and the initial/boundary conditions. For inverse problems, the nondimensional PDEs are supplemented with extra data $[u_m(x, t)]$ and initial/boundary conditions.

07 March 2026 07:10:56

problem, the predicted value must simultaneously satisfy the constraints of the control equations (derivatives calculated by automatic differentiation), boundary conditions, and initial conditions, and a loss function is constructed for optimization, as illustrated in Fig. 1(a). For the inverse problem, the unknown physical parameter (applied voltage in the present study) needs to be included as part of the trainable parameter of the neural network, and the measured data points of the electric field are additionally introduced for the matching loss, and then the network weights and physical parameters are simultaneously updated through multi-objective optimization, as shown in Fig. 1(b).

The architecture of the PINNs for solving forward and inverse problems used in this study is configured as follows: It comprises 8 hidden layers, each with 40 neurons to balance computational precision and training efficiency. The Tanh function is utilized as the nonlinear activation function. Training is performed using an Adam optimizer for 2.5×10^6 iterations, with an initial learning rate of 0.0001. A dynamic learning rate strategy is implemented, where the learning rate is reduced to 90% of its current value every 10 000 steps.

The loss function of PINNs comprises two synergistic components: a data-driven term and a physics-informed term. The data-driven term fits measured data points, and the physics-informed term representing the physical equations in the loss function to train neural networks is very essential to PINNs. These two parts work together to ensure that the neural network not only fits the known data points but also satisfies the given governing equations. The plasma characteristics of plasmas density, electric field, and flux are closely coupled by Eqs. (1)–(4), which presents significant challenges for the training of PINNs. Consequently, these equations are consolidated into unified residual definitions for training PINNs,

$$res_1(x, t) := \frac{\partial \varphi}{\partial x} - E, \quad (8)$$

$$res_2(x, t) := \frac{\partial E}{\partial x} - \frac{q}{\epsilon}(N_i - N_e), \quad (9)$$

$$res_{3,4}(x, t) := \frac{\partial N_{i,e}(x, t)}{\partial t} + \frac{\partial J_{i,e}(x, t)}{\partial x} - S_{i,e}(x, t), \quad (10)$$

$$res_{5,6}(x, t) := J_{i,e}(x, t) \mp \mu_{i,e} E(x, t) N_{i,e}(x, t) - D_{i,e} \frac{\partial N_{i,e}(x, t)}{\partial x}. \quad (11)$$

These residuals can be then used as constraints during the training of the neural networks, in order to encourage them to produce physically consistent predictions. This part of the loss function corresponds to the collocation points. These are points that we randomly choose inside the calculation domains using a latin-hypercube sampling strategy,⁴⁹

$$\mathcal{L}_{residual} = \frac{w_{1,2,3,4,5,6}}{N_{pde}} \sum_{i=1}^{N_{pde}} (res_{1,2,3,4,5,6}(x_i, t_i; \theta))^2, \quad (12)$$

where N_{pde} represents the number of collocation points. w_1, w_2, w_3, w_4, w_5 , and w_6 are the weights for different losses with the fixed value of 1.

In this study, the output of PINNs $\hat{u}(\hat{x}, \hat{t})$ is defined to represent the solution to Eqs. (8), (10), and (11), which is applied to approximate the following mapping:

$$\hat{u}(\hat{x}, \hat{t}) \mapsto [N_e(\hat{x}, \hat{t}), N_i(\hat{x}, \hat{t}), E(\hat{x}, \hat{t}), J_e(\hat{x}, \hat{t}), J_i(\hat{x}, \hat{t})]. \quad (13)$$

This part of the loss function corresponds to fitting the measured simulation data (initial conditions, boundary conditions) in the PINNs, taking the form

$$\begin{aligned} \mathcal{L}_{bc/ic} = & \frac{w_7}{N_\varphi} \sum_{i=1}^{N_\varphi} |\varphi(x_i, t_i) - f(x_i, t_i; \theta)|^2 \\ & + \frac{w_8}{N_{N_e}} \sum_{i=1}^{N_e} |N_e(x_i, t_i) - f(x_i, t_i; \theta)|^2 \\ & + \frac{w_9}{N_{N_i}} \sum_{i=1}^{N_i} |N_i(x_i, t_i) - f(x_i, t_i; \theta)|^2, \end{aligned} \quad (14)$$

where N_φ, N_{N_e} , and N_{N_i} represent the number of measurement points for electric potential, electron density, and ion density, respectively. w_7, w_8 , and w_9 are the weights for different losses with the values of 1, 2, and 2, respectively. The total loss function $\mathcal{L}_{forward}$ of PINNs for solving the forward problem is as follows:

$$\mathcal{L}_{forward} = \mathcal{L}_{residual} + \mathcal{L}_{bc/ic}. \quad (15)$$

The loss function for forward problem enforces dual constraints, which include the physics residual and data label. This goal is to encourage neural networks to find a set of specific parameters θ so that their predictions are consistent with the governing equations and data label, which translates to having the minimum loss function. By satisfying this condition together with matching the measurements at particular points, a model can be obtained, which is capable of inferring the solution at anywhere in the domain for RF discharge.

However, solving the inverse problem by PINNs requires the introduction of additional data as constraints in the loss function. In this paper, the unknown parameter V_0 [applied voltage of Eq. (6)] is taken as an example to show the ability of PINNs for solving the inverse problem. Due to the applied voltage being unknown, the electric field is required as an additional data, usually from experimental measurements for the training of PINNs. The loss function of electric field is as follows:

$$\mathcal{L}_{measurements} = \frac{w_{10}}{N_E} \sum_{i=1}^{N_E} |E(x_i, t_i) - f(x_i, t_i; \theta)|^2, \quad (16)$$

where N_E represents the number of measurement points of the

electric field and w_{10} are the weights for different losses with the value of 1. The total loss function $\mathcal{L}_{inverse}$ for solving the inverse problem is written as

$$\mathcal{L}_{inverse} = \mathcal{L}_{residual} + \mathcal{L}_{bc/ic} + \mathcal{L}_{measurements}. \quad (17)$$

Regarding the inverse problems, although the boundary conditions of potential ($\varphi(0, t)$) are no longer needed, certain electric field data are assumed to be known. At this point, the loss function consists of three parts. One is the loss for governing equation learning and the others are for the IC/BC data of density and potential ($\varphi(d, t)$) and observed data of the electric field. In this study, the applied voltage is the unknown parameter for the training of PINNs. In this way, each field at corresponding positions can be calculated by minimizing the loss function of PINNs, and the unknown parameters values can be retrieved by the well trained network. Finally, the unknown parameter V_0 , potential φ , electric field E , density $N_{e,i}$, and flux $J_{e,i}$ can be obtained simultaneously by minimizing the loss function under physical constraints.

C. Data preparation

Driven by the RF power source, usually the order of magnitude of the plasma parameters, such as applied voltage, electron density, ion density, and electric field, has a very significant relative difference, e.g., $\varphi \sim 10^2 \text{ V}$, $N_e \sim 10^{10} / \text{cm}^{-3}$, $N_i \sim 10^{10} / \text{cm}^{-3}$, $J_e \sim 10^{17} / \text{cm}^{-3}$, $J_i \sim 10^{15} / \text{cm}^{-3}$, and $E \sim 10^4 \text{ V/cm}^{-3}$, which brings substantial difficulty on the training of the neural network with the residual given in Eqs. (8)–(11). In the process of training the neural networks of PINNs, the obvious scale difference significantly influences the magnitude of the gradients backpropagated to calculate the network parameters, accordingly a non-dimensionalization technique is usually applied to re-scale the input and output of the neural networks to a proper value; thus, the PINNs can be more efficiently trained. Based on the characteristic interval and length of the gas discharge system, the following new scales are selected to non-dimensionalize the various physical quantities:^{50,51}

$$\begin{aligned} x_0 &= \frac{1}{Ap}, & E_0 &= Bp, & t_0 &= \frac{x_0}{\mu_e E_0}, \\ N_0 &= \frac{\epsilon_0 E_0}{qx_0}, & J_0 &= \frac{N_0 x_0}{t_0}, & \varphi_0 &= E_0 x_0; \end{aligned} \quad (18)$$

thus, the following non-dimensionalize scales are defined and considered in the loss function of PINNs,

$$\begin{aligned} \hat{x} &= \frac{x}{x_0}, & \hat{t} &= \frac{t}{t_0}, & \hat{\varphi} &= \frac{\varphi}{\varphi_0}, & \hat{E} &= \frac{E}{E_0}, & \hat{N}_e &= \frac{N_e}{N_0}, \\ \hat{N}_i &= \frac{N_i}{N_0}, & \hat{J}_e &= \frac{J_e}{J_0}, & \hat{J}_i &= \frac{J_i}{J_0}. \end{aligned} \quad (19)$$

So, all the output variables are scaled to order $O(1)$ by introducing these new scales, to successfully encourage the training of PINNs coupled with various spatial and temporal lengths. The convergence of PINNs during the training is very essential to ensure the

numerical stability, and thus, the dimensionless potential and electric field should be normalized, which is written as

$$\hat{\varphi} = \frac{\varphi - \varphi_{min}}{\varphi_{max} - \varphi_{min}}, \quad \hat{E} = \frac{E - E_{min}}{E_{max} - E_{min}}, \quad (20)$$

where the normalized potentials φ_{min} and φ_{max} are from the dimensional minimum and maximum values of potential, respectively, and the normalized electric fields E_{min} and E_{max} represent characteristic values obtained by the dimensional minimum/maximum electric fields.

In this study, to validate the accuracy of predictions of PINNs, the absolute errors are introduced to serve as the metric for quantifying the prediction over the computational region. For true values $y_{true}(\hat{x}, \hat{t})$ obtained from discretization methods and predicted values $y_{pred}(\hat{x}, \hat{t})$ given by PINNs, the absolute error between them at each discrete point $\varepsilon_{abs}(\hat{x}, \hat{t})$ is defined as

$$\varepsilon_{abs}(\hat{x}, \hat{t}) = |y_{true}(\hat{x}, \hat{t}) - y_{pred}(\hat{x}, \hat{t})|. \quad (21)$$

Of course, $y_{true}(\hat{x}, \hat{t})$ and $y_{pred}(\hat{x}, \hat{t})$ are dimensionless and normalized values.

For solving the forward problem by PINNs, 10 000 random training data points are generated in the discharge region (for PDE residual). The boundary conditions and initial conditions of electron density and ion density were sampled within one radio frequency cycle after the discharge stabilized, each including 692 sampling points. In addition, only two boundary conditions are given for the potential, namely, the powered electrode with applied RF voltages and grounded electrode with 0, including 492 sampling points in total. For the inverse problem, due to the parameters of the applied voltage being unknown, the applied voltage is unknown and treated as the discharge parameter to be solved based on the extra data of the electric field, which is introduced to be taken as the constraint in the loss function.

III. RESULTS AND DISCUSSION

In this study, the fluid model is used to describe the atmospheric RF plasmas with Helium as the working gas at a gas pressure of 760 Torr. The driven voltage takes a sinusoidal form with the amplitude of 400 V in this simulation. Only direct collision ionization and two kinds of particles (electrons and helium ion) are considered in this model, to validate the ability of PINNs for solving the fluid equations, to clearly show the solving processes. The PINNs for forward and inverse problems were implemented in the PyTorch framework. Trainings were performed on an NVIDIA GeForce RTX 4060 Ti GPU (16 GB VRAM) with the Adam optimizer. Each model converged within approximately 2 h under default hyperparameters. Remarkably, although a trainable boundary parameter (applied voltage) is introduced as extra data in PINNs to solve the inverse problem, the training time of the inverse problem is almost equivalent to that of the forward problem.

A. Results for forward problems

In this subsection, the computational data for solving the forward problem by PINNs are described. Given the applied

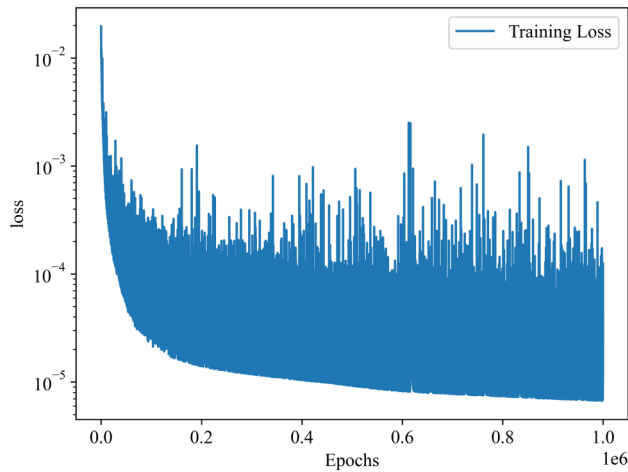


FIG. 2. Loss evolution of PINNs in the forward problem.

voltage as the boundary condition, PINNs can successfully solve the whole fluid model, and the temporal and spatial profiles of electric field, potential, density, and flux are predicted. Compared with traditional discretization methods, the data predicted by PINNs

show good agreement. As shown in Fig. 2, about 100 000 epochs are performed for wholly solving forward problem, which clearly illustrates that loss can be reduced to 10^{-5} during training, indicating that the proposed PINNs for forward problem can reliably predict the plasma characteristics with a given applied voltage. Thus, PINNs provides an alternative tool to solve the fluid model without the requirement of mesh generation.

In this study, a applied voltage of 400 V in atmospheric pressure RF discharge is an example to illustrate the distributions of plasma characteristics by PINNs. Figures 3(a)–3(c) present the temporal–spatial evolution of electron density, ion density, and electric potential, calculated by the traditional method. Figure 3(a) illustrates the spatial profile of electron density, showing the variation in electrons between the two plates driven by the applied voltage. In contrast, the ions are kept in the middle of gas gap due to the large mass in Fig. 3(b). Figure 3(c) illustrates the temporal–spatial evolution of the electric potential. The potential always follows the change of the external applied RF voltage, showing a sinusoidal variation in the anode with the value always being 0 at the cathode. Figures 3(d)–3(e) present the spatiotemporal evolution of electron density, ion density, and electric potential calculated by PINNs. As shown in Figs. 3(d)–3(e), the spatiotemporal evolution characteristics of electron density, ion density, and electric potential exhibit excellent agreement with the reference results in Figs. 3(a)–3(c). This consistency highlights the capability of PINNs to accurately

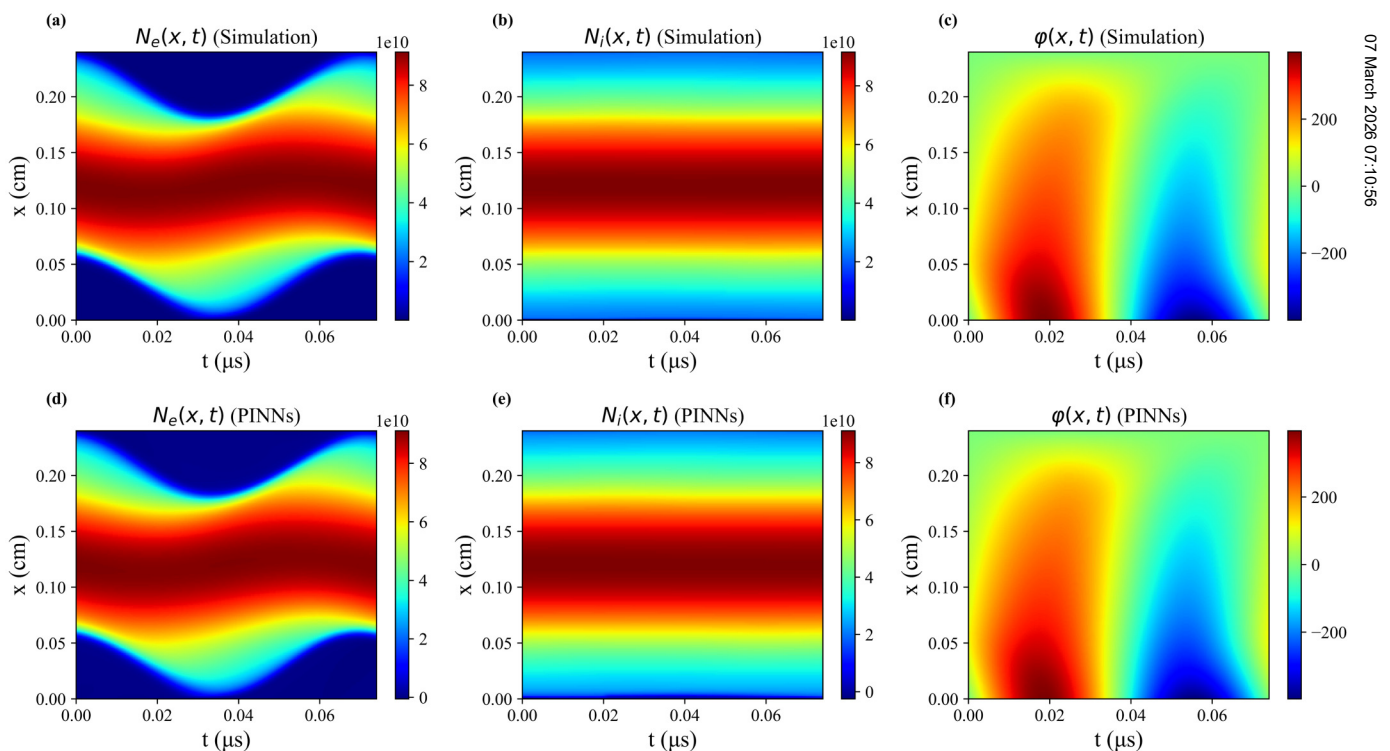


FIG. 3. Temporal–spatial evolution of (a) electron density, (b) ion density, and (c) electric potential solved by the fluid model with traditional method, as well as the (d) electron density, (e) ion density, and (f) electric potential output by PINNs.

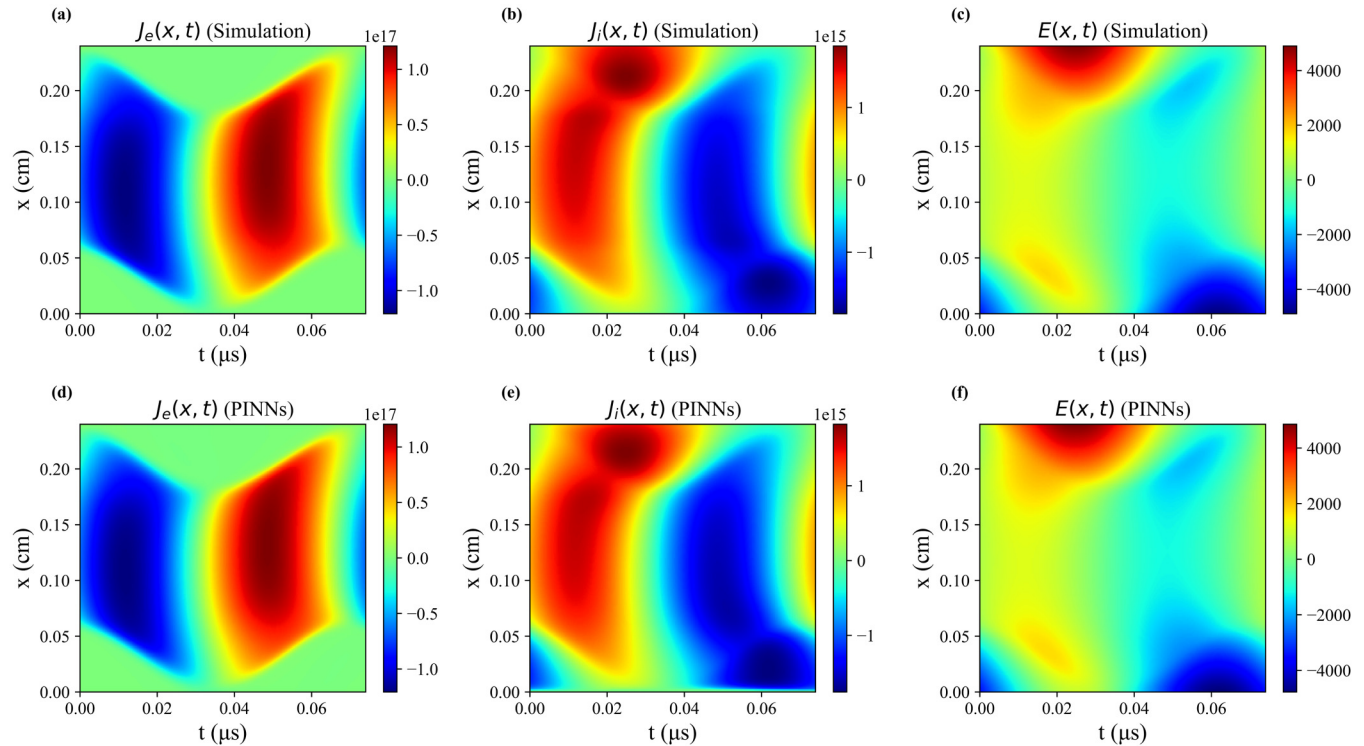


FIG. 4. Temporal-spatial evolution of (a) electron flux, (b) ion flux, and (c) electric field solved by the fluid model with traditional methods, as well as the (d) electron flux, (e) ion flux, and (f) electric field output by PINNs.

capture the complex dynamics of the plasma systems. Additionally, the maximum errors of electron density, ion density, and electric field calculated by the PINNs network are 0.05, 0.08, and 0.007, respectively. There is a strong correlation between the maximum error region and the high gradient region of plasma sheath, which partly reveals the inherent characteristics of PINNs training. In the optimization process, the gradient descent algorithm of the neural network always gives priority to fitting the smooth region (such as the bulk plasma region) and then gradually captures the steep gradient region (such as the sheath region). In the bulk plasma region, the error is basically maintained at about 0.001 with gentle variation in plasma density and electric field, indicating the precision and reliability of PINNs in modeling plasma behavior. The results show that the solution obtained by PINNs is very consistent with the result of the reference fluid model, which accurately captures the key dynamic evolution characteristics of the atmospheric RF discharge.

Figure 4 further illustrates the distribution of electron flux, ion flux, and electric field obtained by the traditional method and PINNs. It can be observed that the change direction of electron flux and ion flux follows the change of polarity of the external RF voltage. Figure 4(c) shows that sheath is a region where the potential changes dramatically, in which electrons are repelled and positive ions are accelerated. When the amplitude of RF voltage changes, the sheath thickness and potential distribution will change accordingly. The errors of flux and electric field remain consistently

low, with their maximum values around 0.02, 0.002, and 0.016, respectively, which demonstrates the accuracy of PINNs in solving nonlinear complex physical processes. The electron flux also reaches its peak value of 1.17×10^{17} . On the other hand, the ions remain almost stationary in the middle of the electrode spacing, and the ion flux with the maximum values of 1.80×10^{15} is significantly smaller than the electron flux. This obvious difference in kinetics between electrons and ions highlights the leading role of electrons in the discharge process, which is consistent with the previous research results.⁵² The above results verify the accuracy of the model in capturing the basic physical aspects of the system.

From Fig. 5(a), it can be observed that there is a distinct plasma sheath region on the right side of the discharge region. In Fig. 5(b), within the sheath region, the electron density rapidly decreases and the ion density decreases more gradually, forming a pronounced sheath region where the electric field is high. In contrast, in the bulk plasma region, the electron density and ion density are nearly equal, and the electric field varies more smoothly. As a result, the electric field strength on the right side of the discharge space rises again. This trend aligns well with previous studies,¹⁰ demonstrating that the well trained PINNs effectively fit the equations of the fluid model and accurately capture the key plasma characteristics of atmospheric pressure RF discharge. The ability of PINNs to reproduce and predict these complex dynamics highlights their potential as a powerful tool for simulating and

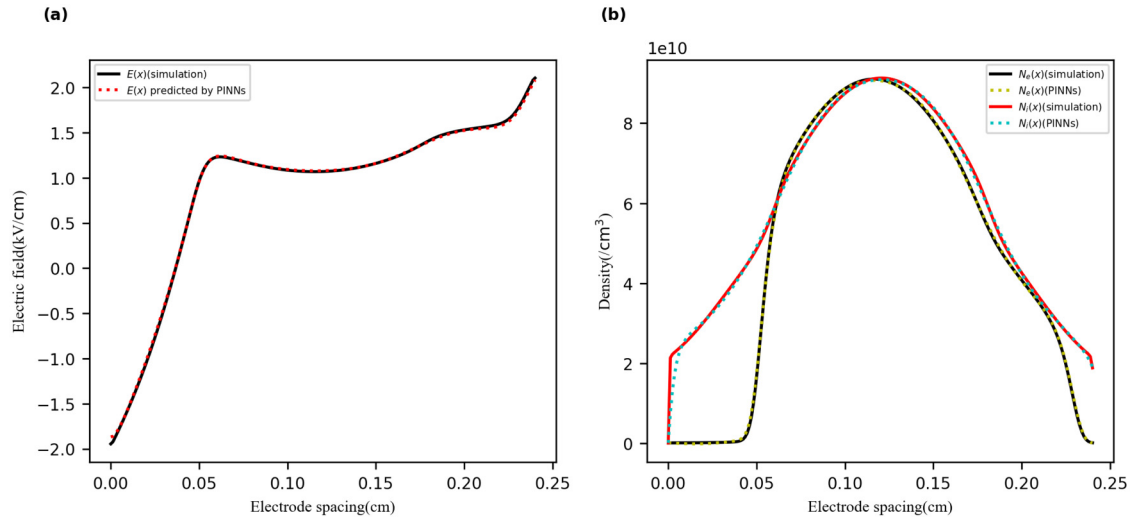


FIG. 5. The spatial distributions of the (a) electric field, (b) electron density, and ion density at the instant when the current density reaches its peak in atmospheric RF discharge under voltages of 400 V, as predicted by PINNs, were compared with fluid simulation results.

analyzing discharge processes, even successfully capturing sharp gradients and dynamic behaviors in the sheath region.

B. Results for inverse problems

In this subsection, the inverse problem for atmospheric RF plasma is solved by PINNs with the aid of experimental and computational data, which cannot be realized by the traditional methods. The amplitude of the applied voltage is regarded as the trainable parameter of PINNs in this study and inferred by introducing additional data of the electric field. Thus, the discharge

parameters can be calculated from the measured or simulated data of plasma characteristics by PINNs.

Figure 6 illustrates the curve of trainable parameter of applied voltage during the inversion process of PINNs. The parameter evolution is constrained solely by initial electric field data at position (d, t) . The initial value of the trainable applied voltage is set to 300 V. During the training process of PINNs, the unknown voltage parameter gradually approaches the final voltage value of 395 V (the relative error of about 1%). The approximation process of the applied voltage suggests that PINNs find the correct answer only after exploring various possibilities through physical constraint optimization. Figure 7 shows a good agreement between the RF

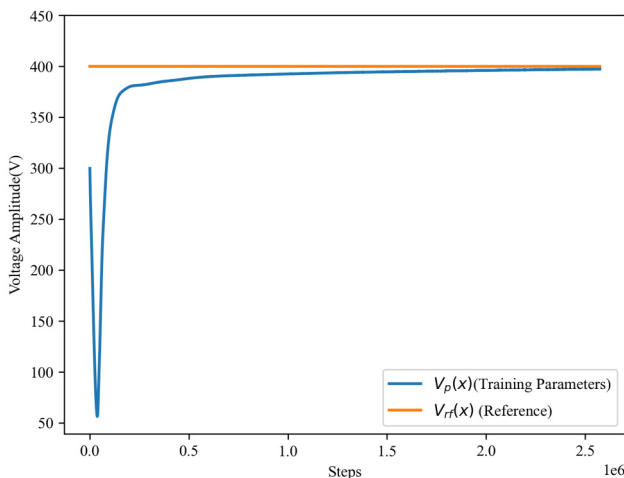


FIG. 6. Comparison between the variation curve of the inverted applied voltage with the training step and the actual results in the PINNs training process.

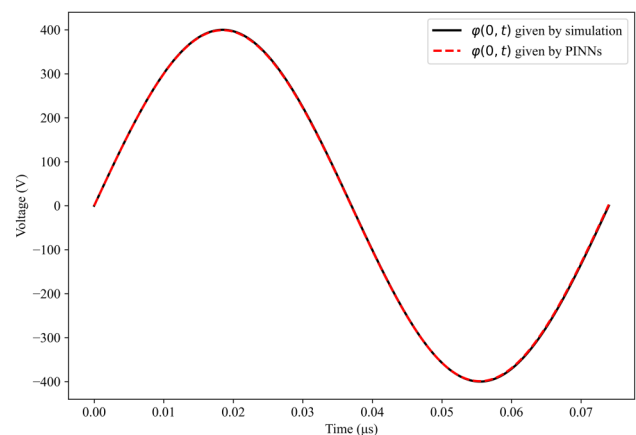
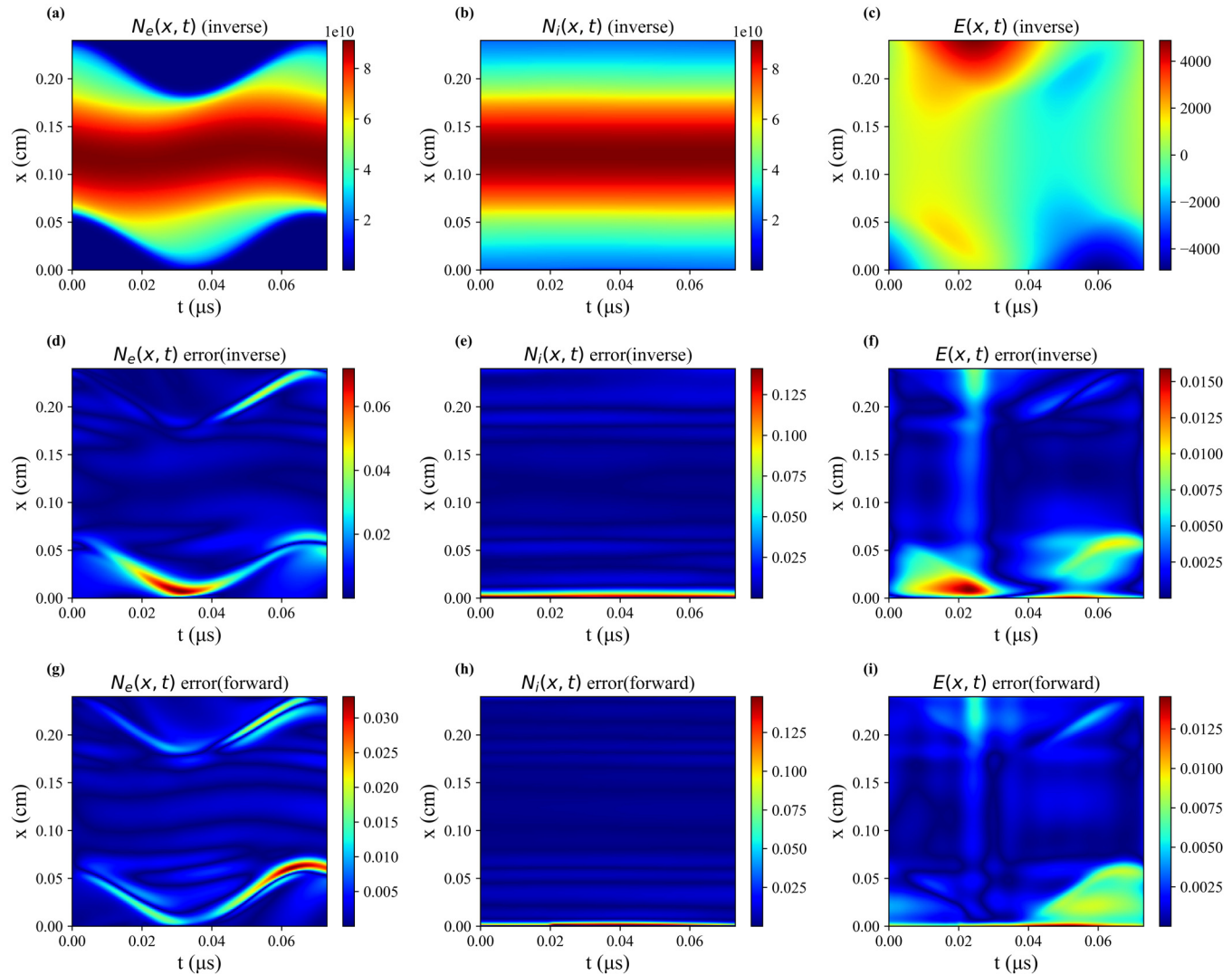


FIG. 7. Comparison of applied voltage calculated by the fluid model and PINNs.

07 March 2026 07:10:56



07 March 2026 07:10:56

FIG. 8. Comparison of plasma parameters from forward/inverse solutions: (a)–(c) Spatiotemporal distributions of electron density, ion density, and electric field from PINNs inverse solution; (d)–(f) the error distributions of forward solution; (g)–(i) are the error distributions of inverse solution.

applied voltage predicted by PINNs and the theoretical value, indicating that the inversion not only captures the correct voltage amplitude but also accurately reconstructs the dynamic behavior of voltage. This highlights how the physical constraints embedded in PINNs can reproduce the whole discharge parameters with extra measured or simulated data.

Figure 8 presents the spatiotemporal distributions of electron density, ion density, and electric field from PINNs for the inverse problem, while (d)–(f) and (g)–(i) compare their error distributions against simulation values for forward and inverse solutions, respectively. The error distributions of electron density and ion density obtained by solving the forward and inverse problems are

consistent. Although the actual physical distribution of electron density are symmetrically periodic, the electron density on the anode sheath is small (about 1/5 of that on the cathode side), and PINNs are not sensitive enough to the training of low-value areas, resulting in an electron density error on the anode side that is about 40 higher than that on the cathode side. For the electric field, the error distribution of the electric field obtained by the forward problem is relatively uniform and mainly concentrated near the anode sheath (the maximum error is about 0.016). Although the electric field obtained by PINNs is constrained by the electric field data $E(d, t)$ at the cathode plate, the error near the cathode is significantly reduced (the error smaller than 0.01), but the error at the

TABLE I. Inversion results and errors for voltage amplitude in different sampling points of the electric field.

Dataset	$E(x_i, t_i)$ sampling points	Inversion voltage amplitude (V)	Relative error(%)
Dataset 1	$E(0, t_i)$	392.17	1.9574
Dataset 2	$E(\frac{1}{4}d, t_i)$	389.12	2.7175
Dataset 3	$E(\frac{1}{2}d, t_i)$	388.85	2.7869
Dataset 4	$E(\frac{3}{4}d, t_i)$	389.03	2.7436
Dataset 5	$E(d, t_i)$	395.27	1.1834
Dataset 6	$E(d, t_i) + E(0, t_i)$	396.61	0.84832

anode (no electric field data constraint) is increased to 0.03, indicating that it is difficult for the equation residual to completely compensate the information loss in the area without measurement data.

This study examines the impact of sampling point locations on the precision of the inverse problem. The analysis employs six groups of electric field sampling points, uniformly distributed along the one-dimensional spatial domain of length d . The measurement locations are specified by the coordinates $(0, t)$, $(\frac{1}{4}d, t)$, $(\frac{1}{2}d, t)$, $(\frac{3}{4}d, t)$, and (d, t) , where t represents time. The initial conditions of the electric field $E(x, 0)$ are also taken as the sampling points. The results of the PINNs inversion under this sampling scheme are presented in Table I, which indicate a pronounced influence of spatial sampling strategy on the accuracy of the reconstructed solution. According to the calculation, only one dataset obtained at the boundaries [$E(0, t)$ and $E(d, t)$] can accurately inverse the applied voltage with the errors of 1.9574 and 1.1834, respectively. When the datasets are obtained at interior positions [$E(d/4, t)$, $E(d/2, t)$, and $E(3d/4, t)$], the inversion errors are relatively larger, reaching 2.7175%, 2.7869%, and 2.7436%, respectively. The multi-position combination [Dataset 6: $E(0, t) + E(d, t)$] reduced the error to 0.84832%, proving that dual-boundary measurements effectively eliminate errors. It is clear that the selection of data points can greatly improve the accuracy of PINNs inversion applied voltage.

The dependence of inversion accuracy in PINNs on the number of sampling points at the given position (d, t) is also

TABLE II. Inversion results and errors for voltage amplitude in different number of sampling points.

Dataset	Number of sampling points	Inversion voltage amplitude (V)	Relative error(%)
Dataset 1	246	395.27	1.1834
Dataset 2	200	395.12	1.2193
Dataset 3	150	395.17	1.2075
Dataset 4	100	395.30	1.1761
Dataset 5	50	395.34	1.164
Dataset 6	30	395.22	1.1959
Dataset 7	20	394.76	1.3101
Dataset 8	10	384.20	3.9495

TABLE III. Inversion results and errors for voltage amplitude in different noise levels of the sampling point.

Dataset	Noise level	Inversion voltage amplitude (V)	Relative error(%)
Dataset 1	0.01	396.63	0.8431
Dataset 2	0.02	397.11	0.7234
Dataset 3	0.04	394.57	1.3582
Dataset 4	0.06	387.54	3.1151
Dataset 5	0.08	383.71	4.0723
Dataset 6	0.10	417.79	4.4475
Dataset 7	0.12	419.27	4.8183
Dataset 8	0.14	422.90	5.7244

investigated as shown in Table II, which gives various numbers of sampling points (246, 200, 150, 100, 50, 30, 20, and 10 data points, respectively). When sampling points decreased from 246 to 30, the inversion error remained stable at 1.16%–1.31%, showing that even with an 87% data reduction, the inversion accuracy does not be seriously affected. However, when the number of points is dropped to 20, the error is sharply increased to 3.95%. Thus, only few number of sampling points are required, and PINNs for inverse problems can accurately infer the applied voltage, which indicates the discharge parameters can be calculated by only a small amount of additional data by PINNs.

The measured data obtained in experiments are always inevitably noisy, which may affect the accuracy of PINNs for inverse problem. In this study, the Gaussian noises with eight different standard deviations (STDs, σ) are added to the measured data of the electric field at position (d, t) with the noise level of 0.01, 0.02, 0.04, 0.06, 0.08, 0.10, 0.12, and 0.14, respectively, as given in Table III. In this study, Gaussian noise is added to the data of the electric field, which is described by two parameters: mean (μ , set to 0) and standard deviation (σ , control noise intensity), so only one parameter (standard deviation σ) is needed to define the noise level. In the data preprocessing stage, the electric field is dimensionless and normalized, so the standard deviation σ range of noise is set to 0.01–0.14 to explore the inversion accuracy of PINNs under different noise levels. To account for the stochastic nature of Gaussian noise, ten independent noise realizations are generated for each noise level, and the final inversion results are obtained by averaging over these ten cases to ensure statistical reliability. When weak Gaussian noise level from 0.01 to 0.02 is introduced to electric field measurements, inversion errors are slightly increased from 0.8431% to 0.7234%, demonstrating the intrinsic robustness of PINNs against noise interference. However, the error increases significantly to 1.3582% at the noise level of 0.04, and the error surges can be observed with peak values at 5.7244% when the noise level is larger than 0.06.

Figure 9 shows the dependence of voltage inversion error on the noise level retrieved from electric field measurement data. The plot demonstrates the robustness of PINNs against measurement noise. The inversion error remains relatively stable at low noise levels ($\sigma \leq 0.04$). However, when the noise level exceeds this threshold, the inversion error increases significantly, and the overall

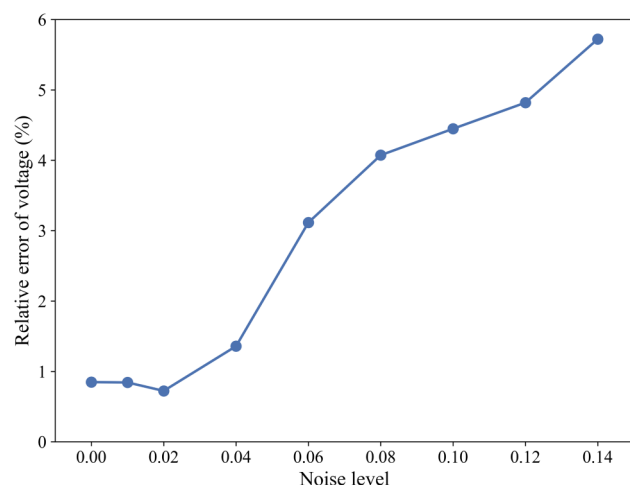


FIG. 9. Inversion voltage error curve with noise level (σ).

inversion error increases with increasing noise. It is also observed that the inversion error at $\sigma = 0.08$ is slightly higher than that at $\sigma = 0.10$. This fluctuation arises from the random nature of Gaussian noise applied in data generation, which may occasionally introduce unfavorable noise realizations that amplify local fitting errors, even when the overall noise intensity is slightly lower. Overall, the calculation shows that the noise (smaller than 0.04) in common experimental measurements does not affect the accuracy of PINNs inversion.^{29,53}

IV. CONCLUSION

In this study, PINNs are successfully applied to solve the forward and inverse problems in atmospheric pressure RF discharges based on fluid description of atmospheric plasmas. For forward problems, PINNs fully obtain the plasma characteristics under given boundary/initial conditions, accurately capturing spatiotemporal profiles of electron/ion densities, electric fields, potentials, and particle fluxes, which demonstrates their capability in solving fluid equations instead of discretization methods. For inverse problems, given the measured or simulated data of the electric field, PINNs achieve high precision inversion of applied voltage in this study ($\sim 1\%$ error). Theoretically, PINNs can infer the discharge parameters, including applied voltage, driving frequency, or electrode spacings from the measured or simulated plasmas characteristics, such as electron density, electric field and electron temperature. The influence of different positions and numbers of sampling points of the electric field on the accuracy of the inversion of discharge parameters is also explored, and the results show that PINNs are able to accurately identify unknown applied voltage even when the experimentally measured data are heavily contaminated by noise. According to the computational data in this study, the ability of PINNs to solve both forward and reverse problems provides possible and potential tools for the intelligent control of atmospheric plasmas in the era of AI.

ACKNOWLEDGMENTS

This work was supported by the Natural Science Foundation of China (No. 12375201).

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Wenkai Li: Data curation (equal); Formal analysis (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal). **Yuantao Zhang:** Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

- 1A. Von Keudell and V. Schulz-Von Der Gathen, *Plasma Sources Sci. Technol.* **26**, 113001 (2017).
- 2X. Zhang, X.-F. Zhang, H.-P. Li, L.-Y. Wang, C. Zhang, X.-H. Xing, and C.-Y. Bao, *Appl. Microbiol. Biotechnol.* **98**, 5387 (2014).
- 3J. Yang, T. Li, C. Zhong, X. Guan, and C. Hu, *J. Electrochem. Soc.* **163**, E288 (2016).
- 4Y. Lu, Z. Ren, H. Yuan, Z. Wang, B. Yu, and J. Chen, *RSC Adv.* **5**, 62619 (2015).
- 5G. Lloyd, G. Friedman, S. Jafri, G. Schultz, A. Fridman, and K. Harding, *Plasma Processes Polym.* **7**, 194 (2010).
- 6J. Park, I. Henins, H. Herrmann, G. Selwyn, and R. Hicks, *J. Appl. Phys.* **89**, 20 (2001).
- 7N. Balcon, G. Hagelaar, and J.-P. Boeuf, *IEEE Trans. Plasma Sci.* **36**, 2782 (2008).
- 8Y. T. Zhang, Q. Q. Li, J. Lou, and Q. M. Li, *Appl. Phys. Lett.* **97**, 141504 (2010).
- 9X. Yuan and L. L. Raja, *IEEE Trans. Plasma Sci.* **31**, 495 (2003).
- 10J. Shi and M. Kong, *J. Appl. Phys.* **97**, 023306 (2005).
- 11X.-L. Wang, S.-H. Gao, and Y.-T. Zhang, *Phys. Plasmas* **28**, 073511 (2021).
- 12Z. Chen, M. Liu, G. Xia, and Y. Huang, *IEEE Trans. Plasma Sci.* **40**, 2861 (2012).
- 13J. J. Munro and J. Tennyson, *J. Vac. Sci. Technol. A* **26**, 865 (2008).
- 14Z. Chen, *The Finite Element Method: Its Fundamentals and Applications in Engineering* (World Scientific, 2011).
- 15C. A. Felippa, *Introduction to Finite Element Methods* (University of Colorado, 2004), Vol. 885.
- 16R. Eymard, T. Gallouët, and R. Herbin, "Finite volume methods," *Handb. Numer. Anal.* **7**, 713–1018 (2000).
- 17A. Fierro, E. Barnat, M. Hopkins, C. Moore, G. Radtke, and B. Yee, *Eur. Phys. J. D* **75**, 151 (2021).
- 18M. M. Turner, *Plasma Processes Polym.* **14**, 1600121 (2017).
- 19P. C. Nguyen, J. B. Choi, H. Udaykumar, and S. Baek, *J. Comput. Inf. Sci. Eng.* **23**, 060808 (2023).
- 20S. Mittal, *J. Syst. Archit.* **104**, 101689 (2020).
- 21M. He, R. Bai, S. Tan, D. Liu, and Y. Zhang, *Plasma Processes Polym.* **21**, 2400020 (2024).
- 22Y.-T. Zhang, S.-H. Gao, and Y.-Y. Zhu, *J. Appl. Phys.* **133**, 053303 (2023).
- 23R. Li, X. Wang, and Y. Zhang, *Appl. Sci.* **14**, 6855 (2024).

07 March 2026 07:10:56

- ²⁴X.-C. Wang and Y.-T. Zhang, *J. Appl. Phys.* **133**, 143301 (2023).
- ²⁵X.-C. Wang, F. Ai, and Y.-T. Zhang, *Phys. Plasmas* **31**, 013504 (2024).
- ²⁶Z.-B. Liu, X.-C. Wang, and Y.-T. Zhang, *IEEE Trans. Plasma Sci.* **51**, 1212 (2023).
- ²⁷L. Zhong, Q. Gu, and B. Wu, *Comput. Phys. Commun.* **257**, 107496 (2020).
- ²⁸R. Anirudh, R. Archibald, M. S. Asif, M. M. Becker, S. Benkadda, P.-T. Bremer, R. H. Budé, C.-S. Chang, L. Chen, R. Churchill et al., *IEEE Trans. Plasma Sci.* **51**, 1750 (2023).
- ²⁹M. Raissi, P. Perdikaris, and G. E. Karniadakis, *J. Comput. Phys.* **378**, 686 (2019).
- ³⁰F. A. Torres, M. M. Negri, M. Nagy-Huber, M. Samarin, and V. Roth, *arXiv:2206.01545* (2022).
- ³¹S. Cuomo, V. S. Di Cola, F. Giampaolo, G. Rozza, M. Raissi, and F. Piccialli, *J. Sci. Comput.* **92**, 88 (2022).
- ³²Z. Hu, K. Shukla, G. E. Karniadakis, and K. Kawaguchi, *Neural Networks* **176**, 106369 (2024).
- ³³I. Depina, S. Jain, S. Mar Valsson, and H. Gotovac, *Georisk: Assess. Manage. Risk Eng. Syst. Geohazards* **16**, 21 (2022).
- ³⁴Y. Chen, L. Lu, G. E. Karniadakis, and L. Dal Negro, *Opt. Express* **28**, 11618 (2020).
- ³⁵L. Lu, R. Pestourie, W. Yao, Z. Wang, F. Verdugo, and S. G. Johnson, *SIAM J. Sci. Comput.* **43**, B1105 (2021).
- ³⁶S. Cai, Z. Wang, F. Fuest, Y. J. Jeon, C. Gray, and G. E. Karniadakis, *J. Fluid Mech.* **915**, A102 (2021).
- ³⁷Z. Mao, A. D. Jagtap, and G. E. Karniadakis, *Comput. Methods Appl. Mech. Eng.* **360**, 112789 (2020).
- ³⁸A. Yazdani, L. Lu, M. Raissi, and G. E. Karniadakis, *PLoS Comput. Biol.* **16**, e1007575 (2020).
- ³⁹E. Haghighat, M. Raissi, A. Moure, H. Gomez, and R. Juanes, *Comput. Methods Appl. Mech. Eng.* **379**, 113741 (2021).
- ⁴⁰F. Sahli Costabal, Y. Yang, P. Perdikaris, D. E. Hurtado, and E. Kuhl, *Front. Phys.* **8**, 42 (2020).
- ⁴¹G. Kissas, Y. Yang, E. Hwuang, W. R. Witschey, J. A. Detre, and P. Perdikaris, *Comput. Methods Appl. Mech. Eng.* **358**, 112623 (2020).
- ⁴²S. Kawaguchi, K. Takahashi, H. Ohkama, and K. Satoh, *Plasma Sources Sci. Technol.* **29**, 025021 (2020).
- ⁴³L. Zhong, B. Wu, and Y. Wang, *Phys. Fluids* **34**, 087116 (2022).
- ⁴⁴S. Kawaguchi, K. Takahashi, and K. Satoh, *J. Phys. D: Appl. Phys.* **56**, 244003 (2023).
- ⁴⁵W.-K. Li and Y.-T. Zhang, *J. Appl. Phys.* **137**, 203304 (2025).
- ⁴⁶Y. T. Zhang, D. Z. Wang, and M. G. Kong, *J. Appl. Phys.* **100**, 063304 (2006).
- ⁴⁷W. Xucheng, G. Shuhan et al., *Plasma Sci. Technol.* **24**, 085401 (2022).
- ⁴⁸L. Yuan, Y.-Q. Ni, X.-Y. Deng, and S. Hao, *J. Comput. Phys.* **462**, 111260 (2022).
- ⁴⁹M. D. McKay, R. J. Beckman, and W. J. Conover, *Technometrics* **42**, 55 (2000).
- ⁵⁰Y. P. Raizer and J. E. Allen, *Gas Discharge Physics* (Springer, 1997), Vol. 2.
- ⁵¹D. D. Šijačić and U. Ebert, *Phys. Rev. E* **66**, 066410 (2002).
- ⁵²J. Shi and M. G. Kong, *Appl. Phys. Lett.* **90**, 111502 (2007).
- ⁵³S. Andreev, A. Bernatskiy, and V. Ochkin, *Vacuum* **180**, 109616 (2020).